
Amazon Rekognition: Image & Video Analysis Powered by AI

Amazon Rekognition is a **deep learning–based computer vision service** from AWS that can **analyze images and videos** to detect objects, people, text, activities, and inappropriate content. It's fully managed, scalable, and easy to integrate into applications.

Core Capabilities of Amazon Rekognition

1. Face Detection & Analysis

- Detect faces in images and videos
 - Extract facial attributes like:
 - Age range
 - Emotions (happy, sad, angry, etc.)
 - Gender
 - Eyes open/closed, beard, glasses, etc.
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2. Face Comparison & Matching

- Compare a face in an image with another (e.g., authentication)
 - Search for a face in a **collection of faces** (1:N match)
 - Useful for **KYC**, **access control**, and **security apps**
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3. Label/Object Detection

- Recognizes **thousands of objects**, scenes, and activities:
 - Cars, animals, people, nature, buildings
 - Actions like "walking", "riding", "playing"
 - Output includes **confidence scores** for each label
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4. Text Detection (OCR)

- Detect and extract text from images (e.g., license plates, signs)
 - Works with both **printed and handwritten** text
 - Multilingual support
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5. Celebrity Recognition

- Automatically identify well-known **celebrities** in images and videos

- Uses AWS-maintained celebrity database
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6. Content Moderation

- Detects **explicit or suggestive content**
 - Nudity, violence, alcohol, drugs, offensive gestures
 - Output provides **confidence levels** for moderation decisions
 - Useful for **user-generated content platforms**
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7. Video Analysis (via Amazon Rekognition Video)

- Real-time and asynchronous video processing
 - Detect faces, objects, celebrities, and activity in videos
 - Use with Amazon Kinesis Video Streams for live feed analysis
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Use Cases

Industry	Example Use Case
 E-commerce	Detect product categories and tags from images
 EdTech	Detect and analyze student engagement from webcam



Security

Facial recognition for building access



Social
Media

Auto-tagging people and detecting unsafe
content



Media

Video metadata tagging and celebrity spotting



Sample CLI/API Usage

```
aws rekognition detect-labels \
```

```
--image '{"S3Object":{"Bucket":"my-bucket","Name":"photo.jpg"}}' \
```

```
--max-labels 5 \
```

```
--min-confidence 80
```

Sample Output:

```
[  
  { "Name": "Person", "Confidence": 99.4 },  
  { "Name": "Car", "Confidence": 88.6 }  
]
```



Security & Compliance

- Data is encrypted **in transit and at rest**
 - **No data stored permanently** unless you opt to save face collections
 - Supports **HIPAA, FedRAMP, GDPR**, etc.
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Pricing Overview

Feature	Pricing Basis
Image analysis	Per image request
Video analysis	Per minute of video
Face collection	Per stored face
First 5,000 images	Free (monthly, for 12 months)

 [AWS Rekognition Pricing](#)

Related AWS Services

- **Amazon S3** – Store and manage images/videos
 - **AWS Lambda** – Trigger analysis on image upload
 - **Amazon Kinesis Video** – Stream video to Rekognition
 - **Amazon Comprehend** – Combine with NLP for enriched analysis
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